Aravind Machiry

Affiliation

Assistant Professor, Department of Electrical and Computer Engineering, Purdue University.

Contact Information

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Research Interests

My research focuses on various aspects of system security, such as vulnerability detection, mobile security, trusted execution environments, static and dynamic analysis of source code, and binaries. I am also interested in developing novel static/dynamic program analysis techniques for system security problems. My research resulted in various Open-source security tools and several Common Vulnerability Exposures (CVEs) in critical system software such as kernel drivers and bootloaders.

Positions & Education

Assistant Professor (PurS3 Lab) — Jan 2021-Present
Department of Electrical and Computer Engineering
Purdue University, West Lafayette, USA

Postdoctoral Researcher — Aug 2020-Dec 2020
University of Pennsylvania, Philadelphia, PA, USA
Advisor: Mayur Naik

Ph.D in Computer Science — Sep 2014- Aug 2020
University of California, Santa Barbara, USA
Advisors: Christopher Kruegel and Giovanni Vigna
Thesis: Securing smart devices from the bottom-up
Supported by:
Symantec Research Labs Graduate Fellowship
UCSB Graduate Division Dissertation Fellowship

Honors & Awards

- **Paul C. Krause Research Award** — 2025
- **Qualcomm Innovation Fellowship** — 2025
- **NSF CAREER Award** — NSF 2024
- **Test of Time Award** for Dynodroid — FSE 2023
- **CSAW Applied Research** Finalist for Argus — CSAW 2023
- **Amazon Research Award** for Securing CI Workflows — 2023
- **Distinguished Paper Award** for 3c — OOPSLA 2022
- **CS Outstanding Dissertation Award** — UCSB 2020
- **CSAW Applied Research** Finalist for DIFUZE — CSAW 2017
- **Internet Defense Prize** Runner up for DR.CHECKER — USENIX Security 2017
- **Distinguished Paper Award** for Ramblr — NDSS 2017
- **Best Paper Award Award** for CLAPP — Grad Workshop 2016
- **Distinguished Artifact Award** for Dynodroid — FSE 2013
- **College of Computing MS Research award** — 2013

Conference Publications

[C.54] G. Farina, R.D. Corte, **Aravind Machiry**, M. Cinque, S. Bagchi. "AnBridge: Protecting On-Device AI with Android Virtualization Framework." *Proceedings of the 56th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (**DSN**)*, **2026**

[C.53] C. Glosner, **Aravind Machiry**. "SoK: All You Ever Wanted to Know About Bootloader Security But Were Afraid to Ask." *Proceedings of the 47th IEEE Symposium on Security and Privacy (**S&P**)*, **2026**

[C.52] G. Tystahl, J. Ghebremichael, S. Muralee, S Cherupattamoolayil, A. Bianchi, **Aravind Machiry**, A. Kapravelos, W. Enck. "Cosseter: GitHub Actions Permission Reduction Using Demand-Driven Static Analysis." *Proceedings of the 47th IEEE Symposium on Security and Privacy (**S&P**)*, **2026**

[C.51] P. Amusuo, O. Cochell, T.V. Lievre, P.V. Patil, **Aravind Machiry**, J.C Davis. "Do Unit Proofs Work? An Empirical Study of Compositional Bounded Model Checking for Memory Safety Verification." *Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (**ICSE**)*, **2026**

[C.50] A. Bhattar, L. Li, M. Zhu, L. Chang, **Aravind Machiry**. "TypeFlexer: Type Directed Flexible Program Partitioning." *Proceedings of the 28th International Symposium on Research in Attacks, Intrusions and Defenses (**RAID**)*, **2025**

[C.49] S.R. Tanksalkar, S. Muralee, S. Danduri, P. Amusuo, A. Bianchi, J.C Davis, **Aravind Machiry**. "LEMIX: Enabling Testing of Embedded Applications as Linux Applications." *Proceedings of the 34th USENIX Security Symposium (**USENIX Security**)*, **2025**

[C.48] X. Wu, **Aravind Machiry** , Yung. Lu, D. Tian. "FlexGE: Towards Secure and Flexible Model Partition for Deep Neural Networks." *Proceedings of the 22nd International Conference on Detection of Intrusions and Malware, and Vulnerability Assessment (**DIMVA**)*, **2025**

[C.47] E. Barsallo Yi, J. Majors, A. Padala, D. Wu, **Aravind Machiry**, S. Bagchi. "Multi-Device Context-Sensitive Attacks Against Privacy." *Proceedings of the 15th ACM Conference on Data and Application Security and Privacy (**CODASPY**)*, **2025**

[C.46] M. Shen, A. Pillai, B. A. Yuan, J. C. Davis, **Aravind Machiry**. "An Empirical Study on the Use of Static Analysis Tools in Open Source Embedded Software." *Proceedings of the 34th ACM SIGSOFT International Symposium on Software Testing and Analysis (**ISSTA**)*, **2025**

[C.45] S. Cherupattamoolayil, C. E. Glosner, A. Bhattar, **Aravind Machiry**. "Adding spatial memory safety to EDK II through checked c." *Proceedings of the 34th ACM SIGSOFT International Symposium on Software Testing and Analysis (**ISSTA**)*, **2025**

[C.44] P. Amusuo, K. A. Robinson, T. Singla, H. Peng, **Aravind Machiry**, S Torres-Arias, L. Simon, J. Davis."ZTDJAVA: Mitigating Software Supply Chain Vulnerabilities via Zero-Trust Dependencies." *Proceedings of the 47th IEEE/ACM International Conference on Software Engineering (**ICSE**)*, **2025**

[C.43] C. Glosner, **Aravind Machiry**. "FuzzUEr: Enabling Fuzzing of UEFI Interfaces." *Proceedings of the ISOC Network and Distributed System Security Symposium (**NDSS**)*, **2025**

[C.42] A. Padala, S. Bagchi, **Aravind Machiry**. "Detection of Device Triggerable Vulnerabilities in Android Companion Apps through Interactive Triaging." *Proceedings 40th ACM/SIGAPP Symposium on Applied Computing (**SAC**)*, **2025**

[C.41] H. Wu, J. Wu, R. Wu, A. Sharma, **Aravind Machiry**, A. Bianchi. "VERIBIN: Adaptive Verification of Patches at the Binary Level." *Proceedings of the ISOC Network and Distributed System Security Symposium (**NDSS**)*, **2025**

[C.40] Hashmat, F., Aljaali, Z.A., Shen, **Aravind Machiry**. "Insights from Running 24 Static Analysis Tools on Open Source Software Repositories." *Proceedings of the Information Systems Security (**ISS**)*, **2024**

[C.39] A Sharma, S Sharma, S Torres-Arias, **Aravind Machiry**. "Rust for Embedded Systems: Current State and Open Problems." *Proceedings of the 31st ACM Conference on Computer and Communications Security (**CCS**)*, **2024**

[C.38] C Zhu, Z Li, A Xue, AP Bajaj, W Gibbs, Y Liu, R Alur, T. Bao, H. Dai, A. Doupé, M. Naik, Y. Shoshitaishvili, R. Wang, **Aravind Machiry**. "TYGR: Type Inference on Stripped Binaries using Graph Neural Networks." *Proceedings of the 32nd USENIX Security Symposium (**USENIX Security**)*, **2024**

[C.37] X Zhang, S Muralee, S Cherupattamoolayil, **Aravind Machiry**. "On the effectiveness of large language models for github workflows." *Proceedings of the 19th International Conference on Availability, Reliability and Security (**ARES**)*, **2024**

[C.36] N. Kumar K, K. Mohan C, **Aravind Machiry**. "Precision Guided Approach to Mitigate Data Poisoning Attacks in Federated Learning." *Proceedings of the 14th ACM Conference on Data and Application Security and Privacy (**CODASPY**)*, **2024**

[C.35] S. Sharma, S.R. Tanksalkar, S. Cherupattamoolayil, **Aravind Machiry**. "Fuzzing API Error Handling Behaviors using Coverage Guided Fault Injection." *Proceedings of the ACM ASIA Conference on Computer and Communications Security (**AsiaCCS**)*, **2024**

[C.34] P. Amusuo, R. Méndez, Z. Xu, **Aravind Machiry**, J. Davis. "Systematically Detecting Packet Validation Vulnerabilities in Embedded Network Stacks." *Proceedings of the 38th ACM International Conference on Automated Software Engineering (**ASE**)*, **2023**

[C.33] J. Srinivasan, R. Tanksalkar, P. Amusuo, J. Davis, **Aravind Machiry**. "Towards rehosting embedded applications as Linux applications." *53rd Annual IEEE/IFIP International Conference on Dependable Systems and Networks-Supplemental Volume (**DSN-Disrupt**)*, **2023**

[C.32] M. Shen, J. Davis, **Aravind Machiry**. "Towards Automated Identification of Layering Violations in Embedded Applications." *Proceedings of the 24th ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems (**LCTES**)*, **2023**

[C.31] S. Muralee, I. Koishybayev, A. Nahapetyan, G. Tystahl, B. Reaves, A. Bianchi, W. Enck, A. Kapravelos, **Aravind Machiry**. "ARGUS: A Framework for Staged Static Taint Analysis of GitHub Workflows and Actions." *Proceedings of the 31st USENIX Security Symposium (**USENIX Security**)*, **2023**

[C.30] J. Majors, E. Barsallo Yi, A. Maji, D. Wu, S. Bagchi, **Aravind Machiry**. "Security Properties of Virtual Remotes and SPOOKing their violations." *Proceedings of the ACM ASIA Conference on Computer and Communications Security (**AsiaCCS**)*, **2023**

[C.29] M. Busch, M. Payer, **Aravind Machiry**, C. Kruegel, G. Vigna, C. Spensky. "TEEzz: Fuzzing Trusted Applications on COTS Android Devices." *Proceedings of the 44nd IEEE Symposium on Security and Privacy (**S&P**)*, **2023**

[C.28] V. Singhal, A. Pillai, C. Saumya, M. Kulkarni, **Aravind Machiry**. "Cornucopia: A Framework for Feedback Guided Generation of Binaries." *Proceedings of the 37th ACM International Conference on Automated Software Engineering (**ASE**)*, **2022**

[C.27] I. Koishybayev, A. Nahapetyan, R. Zachariah, S. Muralee, B. Reaves, A. Kapravelos, **Aravind Machiry**. "Characterizing the Security of Github CI Workflows." *Proceedings of the 31st USENIX Security Symposium (**USENIX Security**)*, **2022**

[C.26] D. Das, P. Bose, **Aravind Machiry**, S. Mariani, Y. Shoshitaishvili, C. Kruegel and G. Vigna. "Hybrid Pruning: Towards A Precise Static Analysis." *Proceedings of the 16th International Conference on Detection of Intrusions and Malware, and Vulnerability Assessment (**DIMVA**)*, **2022**

[C.25] P. Pashakhanloo, **Aravind Machiry**, H. Choi, A. Canino, K. Heo, I. Lee, M. Naik. "PacJam: Securing Dependencies Continuously via Package-Oriented Debloating." *Proceedings of the ACM ASIA Conference on Computer and Communications Security (**AsiaCCS**)*, **2022**

[C.24] **Aravind Machiry**, J. Kastner, M. McCutchen, A. Eline, K. Headley, M. Hicks. "C to Checked C by 3C." *Proceedings of the Object-oriented Programming, Systems, Languages, and Applications (**OOPSLA**)*, **2022**. Won **Distinguished Paper Award**.

[C.23] C. Garg, **Aravind Machiry**, A. Continella, C. Kruegel, and G. Vigna. "Toward a Secure Crowdsourced Location Tracking System." *Proceedings of the ACM Conference on Security and Privacy in Wireless and Mobile Networks (**WiSec**)*, **2021**

[C.22] Z. Li, **Aravind Machiry**, B. Chen, M. Naik, K. Wang, and L. Song. "ARBITRAR: User-Guided API Misuse Detection." *Proceedings of the 42nd IEEE Symposium on Security and Privacy (**S&P**)*, **2021**

[C.21] C. Spensky, **Aravind Machiry**, N. Burow, H. Okhravi, R. Housley, Z. Gu, H. Jamjoom, C. Kruegel, and G. Vigna. "Glitching Demystified: Analyzing Control-flow-based Glitching Attacks and Defenses." *Proceedings of the 51st International Conference on Dependable Systems and Networks (**DSN**)*, **2021**

[C.20] N. Redini, A. Continella, D. Das, G.De Pasquale, N. Spahn, **Aravind Machiry**, A. Bianchi, C. Kruegel, and G. Vigna. "DIANE: Identifying Fuzzing Triggers in Apps to Generate Under-constrained Inputs for IoT Devices." *Proceedings of the 42nd IEEE Symposium on Security and Privacy (**S&P**)*, **2021**

[C.19] D. Meng, M. Guerriero, **Aravind Machiry**, H. Aghakhani, P. Bose, A. Continella, C. Kruegel and G. Vigna. "Bran: Reduce Vulnerability Search Space in Large Open Source Repositories by Learning Bug Symptoms." *Proceedings of the ACM ASIA Conference on Computer and Communications Security (**AsiaCCS**)*, **2021**

[C.18] C. Spensky, **Aravind Machiry**, N. Redini, C. Unger, G. Foster, E. Balsband, H. Okhravi, C. Kruegel and G. Vigna. "Conware: Automated Modeling of Hardware Peripherals." *Proceedings of the ACM ASIA Conference on Computer and Communications Security (**AsiaCCS**)*, **2021**

[C.17] C. Salls, **Aravind Machiry**, A. Doupe, Y. Shoshitaishvili, C. Kruegel, and G. Vigna. "Exploring Abstraction Functions in Fuzzing." *Proceedings of the 2020 IEEE Conference on Communications and Network Security (**CNS**)*, **2020**

[C.16] C. Spensky, **Aravind Machiry**, M. Busch, K. Leach, R. Housley, C. Kruegel, and G. Vigna. "TRUST.IO: Protecting Physical Interfaces on Cyber-physical Systems." *Proceedings of the 2020 IEEE Conference on Communications and Network Security (**CNS**)*, **2020**

[C.15] **Aravind Machiry**, N. Redini, E. Cammellini, C. Kruegel and G. Vigna. "SPIDER: Enabling Fast Patch Propagation in Related Software Repositories." *Proceedings of the 41st IEEE Symposium on Security and Privacy (**S&P**)*, **2020**

[C.14] N. Redini, **Aravind Machiry**, R. Wang, C. Spensky, A. Continella Y. Shoshitaishvili, C. Kruegel and G. Vigna. "KARONTE: Detecting Insecure Multi-binary Interactions in Embedded Firmware." *Proceedings of the 41st IEEE Symposium on Security and Privacy (**S&P**)*, **2020**

[C.13] E. Gustafson, M. Muench, C. Spensky, N. Redini, **Aravind Machiry**, Y. Fratantonio, D. Balzarotti, A. Francillon, Y. E. Choe, C. Kruegel, G. Vigna. "Toward the Analysis of Embedded Firmware through Automated Re-hosting." *Proceedings of the 22nd International Symposium on Research in Attacks, Intrusions and Defenses (**RAID**)*, **2019**

[C.12] N. Redini, R. Wang, **Aravind Machiry**, Y. Shoshitaishvili, C. Kruegel and G. Vigna. "BinTrimmer: Towards Static Binary Debloating Through Abstract Interpretation." *Proceedings of the 16th International Conference on Detection of Intrusions and Malware, and Vulnerability Assessment (**DIMVA**)*, **2019**

[C.11] **Aravind Machiry**, N. Redini, E. Gustafson, Y. Fratantonio, Y. E. Choe, C. Kruegel and G. Vigna. "Using Loops For Malware Classification Resilient to Feature-unaware Perturbations." *Proceedings of the 34th Annual Application Security Application Conference (**ACSAC**)*, **2018**

[C.10] A. Bianchi, Y. Fratantonio, **Aravind Machiry**, C. Kruegel, G. Vigna, S. Chung, W. Lee. "Broken Fingers: On the Usage of the Fingerprint API in Android." *Proceedings of the ISOC Network and Distributed System Security Symposium (**NDSS**)*, **2018**.

[C.9] N. Redini, **Aravind Machiry**, D. Das, Y. Fratantonio, A. Bianchi, E. Gustafson, Y. Shoshitaishvili, C. Kruegel, G. Vigna. "BootStomp: On the Security of Bootloaders in Mobile Devices." *Chaos Communication Congress (**34C3**)*, **2017**.

[C.8] J. Corina, **Aravind Machiry**, C. Salls, Y. Shoshitaishvili, Shuang Hao, C. Kruegel, and G. Vigna. "DIFUZZING Android Kernel Drivers." *Black Hat Europe London, UK December (**BH EU**)*, **2017**.

[C.7] J. Corina, **Aravind Machiry**, C. Salls, Y. Shoshitaishvili, Shuang Hao, C. Kruegel, and G. Vigna. "DIFUZE: Interface Aware Fuzzing for Kernel Drivers." *Proceedings of the 24th ACM Conference on Computer and Communications Security (**CCS**)*, **2017**. Finalist for **CSAW Applied Research Competition**.

[C.6] **Aravind Machiry**, C. Spensky, J. Corina, N. Stephens, C. Kruegel, G. Vigna. "DR.CHECKER: A Soundy Analysis for Linux Kernel Drivers." *Proceedings of the 26th USENIX Security Symposium (**USENIX Security**)*, **2017**. Runner up for **Facebook Internet Defense Prize**

[C.5] N. Redini, **Aravind Machiry**, D. Das, Y. Fratantonio, A. Bianchi, E. Gustafson, Y. Shoshitaishvili, C. Kruegel, G. Vigna. "BootStomp: On the Security of Bootloaders in Mobile Devices." *Proceedings of the 26th USENIX Security Symposium (**USENIX Security**)*, **2017**.

[C.4] **Aravind Machiry**, E. Gustafson, C. Spensky, C. Salls, N. D. Stephens, R. Wang, A. Bianchi, Y. E. Choe,

C. Kruegel, G. Vigna. "BOOMERANG: Exploiting the Semantic Gap in Trusted Execution Environments." *Proceedings of the ISOC Network and Distributed System Security Symposium (**NDSS**)*, **2017**.

[C.3] R. Wang, Y. Shoshitaishvili, A. Bianchi, **Aravind Machiry**, J. Grosen, P. Grosen, C. Kruegel, G. Vigna. "Ramblr: Making Reassembly Great Again." *Proceedings of the ISOC Network and Distributed System Security Symposium (**NDSS**)*, **2017**. Won **Distinguished Paper Award**.

[C.2] Y. Fratantonio, **Aravind Machiry**, A. Bianchi, C. Kruegel, G. Vigna. "CLAPP: Characterizing Loops in Android Applications." *Proceedings of the ACM Symposium on Foundations of Software Engineering (**FSE**)*, **2015**.

[C.1] **Aravind Machiry**, R. Tahiliani, M. Naik. "Dynodroid: An Input Generation System for Android Apps." *Proceedings of the ACM Symposium on Foundations of Software Engineering (**FSE**)*, **2013**. Won **Distinguished Artifact Award**.

Workshop Publications

[W.4] D. Quarta, M. Ianni, **Aravind Machiry**, Y. Fratantonio, E. Gustafson, D. Balzarotti, M. Lindorfer, C. Kruegel, and G. Vigna. "Tarnhelm: Isolated, Transparent and Confidential Execution of Arbitrary Code in ARM's TrustZone." *Proceedings of the ACM Workshop on Research on Offensive and Defensive Techniques in the Context of Man At The End Attacks (**CheckMate**)*, **2021**

[W.3] **Aravind Machiry**, N. Redini, E. Gustafson, H. Aghakhani, C. Kruegel and G. Vigna. "Detecting Deceptive Reviews using Generative Adversarial Networks." *Proceedings of the 2nd Binary Analysis Research Workshop (**BAR**)*, **2019**.

[W.2] H. Aghakhani, **Aravind Machiry**, S. Nilizadeh, C. Kruegel and G. Vigna. "Detecting Deceptive Reviews using Generative Adversarial Networks." *Proceedings of the 1st Deep Learning and Security Workshop (**DLS**)*, **2018**.

[W.1] Y. Fratantonio, **Aravind Machiry**, A. Bianchi, C. Kruegel, G. Vigna. "CLAPP: Characterizing Loops in Android Applications (Invited Talk)." *Proceedings of the International Workshop on Software Development Lifecycle for Mobile (**DeMobile**)*, **2015**.

Magazine Publications

[M.2] A. Bianchi, K. Borgolte, J. Corbetta, F. Disperati, A. Dutcher, J. Grosen, P. Grosen, **Aravind Machiry**, C. Salls, N. Stephens, G. Vigna, R. Wang (Authors listed alphabetically). "Mechanical Phish: Resilient Autonomous Hacking." *IEEE Security & Privacy Magazine - **SPSI: Hacking without Humans 2018***.

[M.1] A. Bianchi, K. Borgolte, J. Corbetta, F. Disperati, A. Dutcher, J. Grosen, P. Grosen, **Aravind Machiry**, C. Salls, N. Stephens, G. Vigna, R. Wang (Authors listed alphabetically). "Cyber Grand Shellphish." ***Phrack***, **2017**.

Posters

[P.2] Shashank Sharma, Ayushi Sharma, **Aravind Machiry**. "(POSTER) Aunor: Converting Rust crates to [no_std] at scale." *Proceedings of the 14th ACM Conference on Data and Application Security and Privacy (**CODASPY**)*, **2024**

[P.1] **Aravind Machiry**, H. Touma, R. Chen, M. Hicks. "(POSTER) Automated conversion of legacy code to Checked C." *Proceedings of the IEEE Secure Development Conference (**SecDev**)*, **2019**

Talks

- Unleashing D on Android Kernel Drivers — Nullcon 2018
- Piston: Uncooperative Remote Runtime Patching — ACSAC 2018
- Cyber Grand Shellphish — DEFCON, USA, 2016
- Million Dollar Baby: Towards ANGRly conquering DARPA CGC — Nullcon 2016

Professional Activities

Technical Program Committe Member

Conferences (under-approximated):

- **2026:** USENIX Security, IEEE S&P, ACM CCS, ISOC NDSS, RAID
- **2025:** ISOC NDSS, USENIX Security, ACM CCS, OOPSLA, IEEE S&P, USENIX ATC, ACM ACSAC

- **2024:** IEEE S&P, ACM CCS, USENIX Security, USENIX ATC, ACM ACSAC, ISOC NDSS, DIMVA, RAID, USENIX WOOT
- **2023:** ACM CCS, ACSAC, SecDev, AsiaCCS, WOOT, EuroSec, ESORICS
- **2022:** AsiaCCS, Middleware, RAID, WOOT, EuroSec

Organizing Member (Chair)

Conferences:

ISOC NDSS (Associate Chair - 2026)

Workshops:

BAR Workshop (Co-located with NDSS) (2023, 2022)

Teaching

ECE 26400 - Advanced C Programming, Purdue University Sp'2023
ECE 46900 - Operating Systems Engineering, Purdue University Sp'2026, Sp'2025, Sp'2024, Sp'2022, Sp'2021
ECE 69500 - Holistic Software Security, Purdue University Fa'2025, Fa'2024, Fa'2023, Fa'2022, Fa'2021